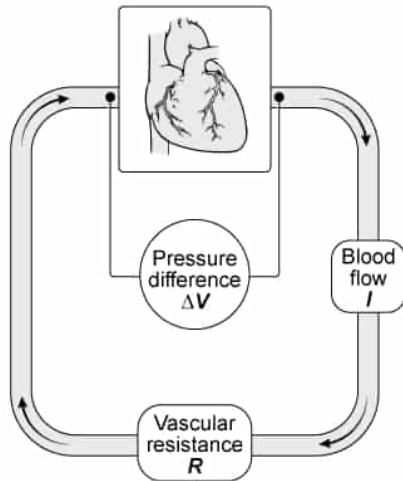


In hemodynamics, blood flow through the cardiovascular system can be modeled as an electric circuit in which the blood serves as electricity, the blood vessels as resistive wires, and the heart as a battery (Figure 1).



**Figure 1** Cardiovascular circuit model

Ohm's law states that the voltage drop  $\Delta V$  across each element, the current  $I$  flowing through it, and its electrical resistance  $R$  are related by  $\Delta V = IR$ . In a blood vessel, the pressure difference  $\Delta P$  between one vessel and the next replaces  $\Delta V$ , the volumetric blood flow  $Q$  replaces  $I$ , and the vascular resistance  $R$  replaces electrical resistance. Therefore, Ohm's law for blood flow in a vessel is

$$\Delta P = QR$$

**Equation 1**

Vascular resistance is due to the blood's viscosity  $\eta$  and the dimensions of the vessel through which it flows. Assuming blood vessels are cylindrical,  $R$  can be approximated as

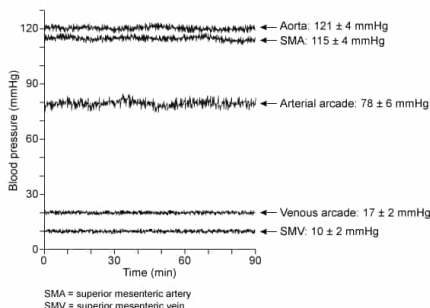
$$R = \frac{8\eta L}{\pi r^4}$$

**Equation 2**

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

In a study of the circulatory system in rats, researchers measured the intravascular blood pressure of the mesenteric blood vessels (the mesentery is the set of tissues that holds the intestines in place). Blood flows in the following order: aorta, superior mesenteric artery (SMA), arterial arcade, venous arcade, superior mesenteric vein (SMV).

For the procedure, catheters with pressure transducers were inserted into each vessel via laparotomy, an invasive procedure that involves a surgical incision into the abdominal cavity. Systolic blood pressure measurements were taken simultaneously at the proximal end (beginning) of each blood vessel.



**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

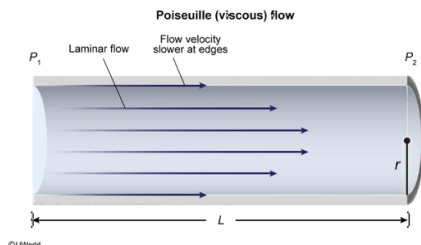
Which change would result in the greatest decrease in the volumetric blood flow rate within a mesenteric vessel?

- ☐ A. Decrease the viscosity of the blood by a factor of 4 [8%]
- ☐ B. Increase the length of the vessel by a factor of 9 [26%]
- ✓ ☒ C. Decrease the radius of the vessel by a factor of 2 [59%]
- ☐ D. Increase the pressure difference by a factor of 6 [6%]

Correct

58% answered correctly

### Explanation:



**Poiseuille's law** describes the laminar flow (ie, no turbulence) of a viscous, incompressible **fluid through a pipe**. The **viscosity** of a fluid measures the internal frictional force that resists the flow. A fluid is incompressible if its density and volume do not change appreciably due to changes in pressure. As stated in the passage, the combination of Equation 1 and Equation 2 is Poiseuille's law, which can be used to **determine the blood flow rate**:

$$Q = \frac{\Delta P}{R} = \frac{\pi r^4 \Delta P}{8\eta L}$$

The **greatest decrease** in the blood flow rate is achieved when the **radius is reduced** by a factor of 2. Poiseuille's law has a **fourth-power dependence** on  $r$  and a linear (first-power) dependence on all the other variables, so the change in radius yields:

$$\left(\frac{r}{2}\right)^4 = \frac{r^4}{2^4} = \frac{r^4}{16}$$

Therefore, the blood flow rate  $Q$  decreases by a factor of 16 when the vessel's radius  $r$  is halved.

**(Choices A and D)** A decrease in viscosity increases the blood flow rate because the flow rate is inversely proportional to viscosity. Similarly, an increase in the blood pressure difference increases the blood flow rate because they are directly proportional to each other.

**(Choices B)** An increase in vessel length reduces the blood flow rate by a factor of 9, but this reduction is less than the one produced by a decrease in the vessel radius.

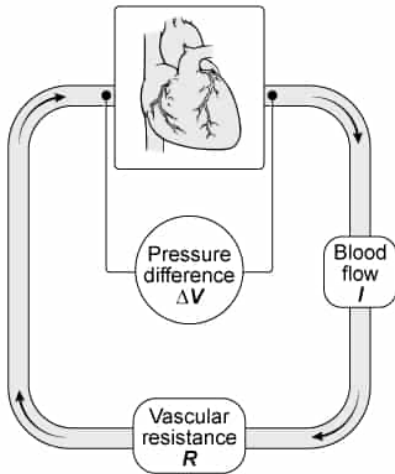
### Educational objective:

Poiseuille's law is used to model fluid flow in pipes, assuming laminar flow of viscous and incompressible fluids. According to Poiseuille's law, the flow rate is directly proportional to vessel radius and pressure difference, and inversely proportional to viscosity and vessel length.

Time spent: 0 sec  
Subject:  
Physics

QID: 400214  
System:  
Electrostatics and Circuits

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**Figure 1** Cardiovascular circuit model

Ohm's law states that the voltage drop  $\Delta V$  across each element, the current  $I$  flowing through it, and its electrical resistance  $R$  are related by  $\Delta V = IR$ . In a blood vessel, the pressure difference  $\Delta P$  between one vessel and the next replaces  $\Delta V$ , the volumetric blood flow  $Q$  replaces  $I$ , and the vascular resistance  $R$  replaces electrical resistance. Therefore, Ohm's law for blood flow in a vessel is

$$\Delta P = QR$$

**Equation 1**

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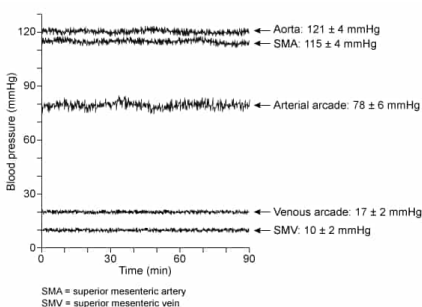
$$R = \frac{8\eta L}{\pi r^4}$$

**Equation 2**

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

In a study of the circulatory system in rats, researchers measured the intravascular blood pressure of the mesenteric blood vessels (the mesentery is the set of tissues that holds the intestines in place). Blood flows in the following order: aorta, superior mesenteric artery (SMA), arterial arcade, venous arcade, superior mesenteric vein (SMV).

For the procedure, catheters with pressure transducers were inserted into each vessel via laparotomy, an invasive procedure that involves a surgical incision into the abdominal cavity. Systolic blood pressure measurements were taken simultaneously at the proximal end (beginning) of each blood vessel.



**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

Which of the following can be used as units for  $\eta$  in Equation 2?

- ☐ A.  $\text{Pa} \cdot \text{m}^2 \cdot \text{s}$  [20%]
- ☐ B.  $\frac{\text{Pa} \cdot \text{m}^2}{\text{s}}$  [38%]
- ✓ ☒ C.  $\text{Pa} \cdot \text{s}$  [34%]
- ☐ D.  $\frac{\text{Pa}}{\text{s}}$  [7%]

Correct

34% answered correctly

### Explanation:

The **units** of a variable in an equation can be **determined from** the units of the **other variables**. Viscosity  $\eta$  is used in Equation 2; however, the units of vascular resistance  $R$  are unknown. Because the units of  $R$  can be determined from Equation 1, these two equations can be combined to **eliminate  $R$** :

$$\frac{8\eta L}{\pi r^4} = \frac{\Delta P}{Q}$$

To determine the units of  $\eta$ , the other variables are rewritten with their **SI units** of length  $L$  and radius  $r$  in meters (m), pressure change  $\Delta P$  in pascals (Pa), and volumetric flow rate  $Q$  in cubic meters per second ( $\text{m}^3/\text{s}$ ). Note that 8 and  $\pi$  are dimensionless quantities (ie, they have no units).

$$\frac{\eta[\text{m}]}{[\text{m}^4]} = \frac{[\text{Pa}]}{[\text{m}^3]/[\text{s}]}$$

Solving for  $\eta$  and **simplifying**:

$$\eta = \frac{[\text{Pa}] \cdot [\text{m}^4] \cdot [\text{s}]}{[\text{m}^3] \cdot [\text{m}]}$$

$$\eta = \text{Pa} \cdot \text{s}$$

**(Choice A)** The units for flow velocity  $v$  in  $\text{m}/\text{s}$  were used in calculating this result instead of the correct units for volumetric flow rate  $Q$ .

**(Choices B and D)** The unit  $1/(1/[\text{s}])$  was incorrectly simplified to  $1/\text{s}$  or the expression  $\text{m}^3 \cdot \text{s}$  was incorrectly used as the unit for volumetric flow rate  $Q$  when calculating these results.

### Educational objective:

The units of a variable in an equation can be determined by rewriting the other variables using their units and solving for the units of the target variable.

Time spent: 0 sec

Subject:

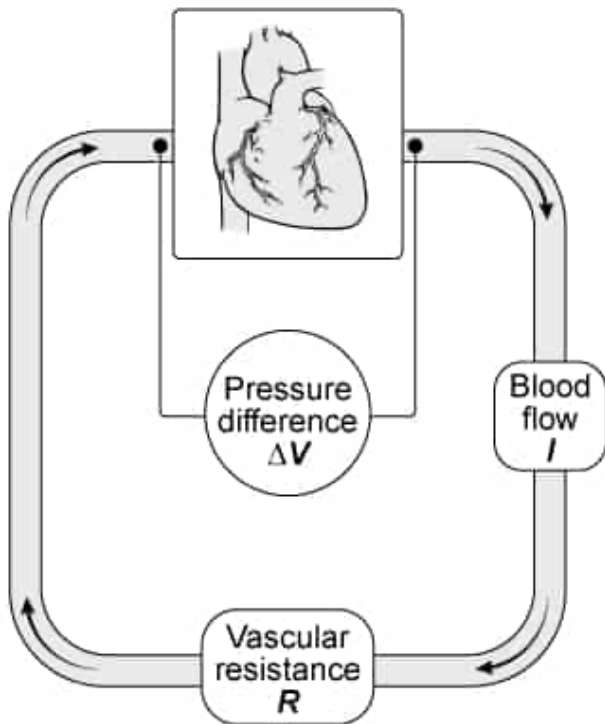
Physics

QID:400215

System:

Electrostatics and Circuits

In hemodynamics, blood flow through the cardiovascular system can be modeled as an electric circuit in which the blood serves as electricity, the blood vessels as resistive wires, and the heart as a battery (Figure 1).



**Figure 1** Cardiovascular circuit model

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$$\Delta P = QR$$

**Equation 1**

Vascular resistance is due to the blood's viscosity  $\eta$  and the dimensions of the vessel through which it flows. Assuming blood vessels are cylindrical,  $R$  can be approximated as

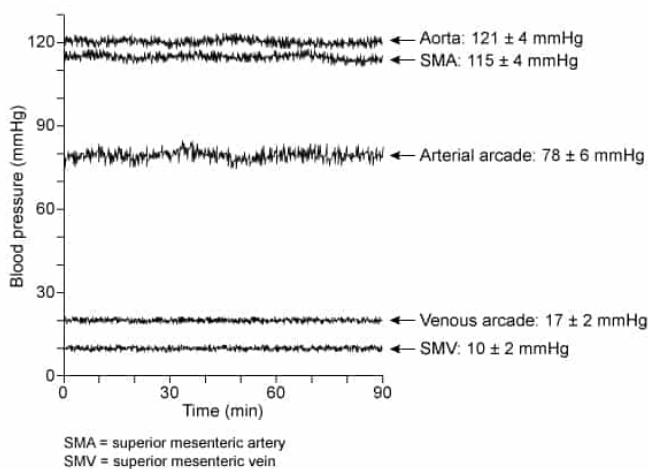
$$R = \frac{8\eta L}{\pi r^4}$$

**Equation 2**

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

In a study of the circulatory system in rats, researchers measured the intravascular blood pressure of the mesenteric blood vessels (the mesentery is the set of tissues that holds the intestines in place). Blood flows in the following order: aorta, superior mesenteric artery (SMA), arterial arcade, venous arcade, superior mesenteric vein (SMV).

For the procedure, catheters with pressure transducers were inserted into each vessel via laparotomy, an invasive procedure that involves a surgical incision into the abdominal cavity. Systolic blood pressure measurements were taken simultaneously at the proximal end (beginning) of each blood vessel.



**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

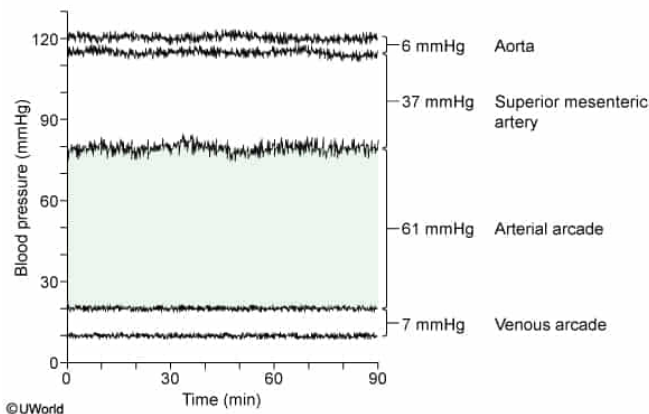
Given the relationship between vascular resistance and blood pressure decay in Equation 1, which of the following blood vessel categories has the greatest total vascular resistance during systole? (Note: Vessel arcades are comprised of multiple individual blood vessels.)

- ☐ A. Aorta [49%]
- ☒ B. Arterial arcade [29%]
- ☐ C. Venous arcade [16%]
- ☐ D. Superior mesenteric artery [5%]

Correct

28% answered correctly

## Explanation:



Vascular resistance is the resistance in a blood vessel that must be overcome for blood to flow. Equation 1 can be **rearranged** to  $R = \Delta P / Q$ , showing resistance  $R$  is **directly proportional** to pressure difference  $\Delta P$  and **inversely proportional** to flow rate  $Q$ .  $Q$  must be the same for each vessel because the continuity equation states that the same volume of blood per unit time must flow through all the vessels that are connected in parallel. Thus,  **$R$  is dependent only on  $\Delta P$** , and the vessel with the greatest  $\Delta P$  has the greatest  $R$ .

All pressure measurements in the study were recorded during systole, the point of greatest pressure. Therefore, the **pressure drop** across a vessel is the **difference** between its **pressure and the pressure of the next vessel**. These pressure differences are calculated using Figure 2. The systolic pressure at the beginning of the arterial arcade is 78 mmHg and it drops to 17 mmHg at the next vessel, the venous arcade, and  $\Delta P$  for the arterial arcade is 61 mmHg, which is greater than  $\Delta P$  for any other vessel. Therefore, the arterial arcade has the greatest  $R$ .

**(Choice A)** Although the aorta has the highest pressure,  $\Delta P$  between the aorta and the next vessel, the SMA, is only 6 mmHg. Therefore, the aorta has

a lower  $R$  than the arterial arcade.

**(Choice C)**  $\Delta P$  between the venous arcade and the next vessel, the SMV, is only 7 mmHg. Therefore, the venous arcade does not have the greatest  $R$ .

**(Choice D)**  $\Delta P$  between the SMA and the next vessel, the arterial arcade, is only 37 mmHg. Therefore, the SMA does not have the greatest  $R$ .

**Educational objective:**

The vascular resistance of a vessel is directly proportional to the pressure difference and inversely proportional to the volumetric flow rate. When the volumetric blood flow rate is the same in two vessels, the vessel with the greater pressure difference has the greater resistance to flow.

**Time spent:**0 sec

**Subject:**

Physics

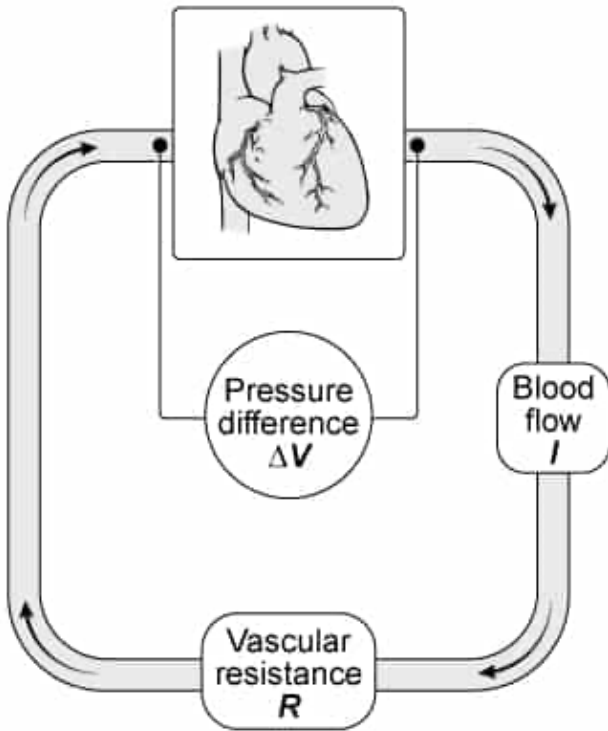
**QID:**400216

**System:**

Electrostatics and Circuits



In hemodynamics, blood flow through the cardiovascular system can be modeled as an electric circuit in which the blood serves as electricity, the blood vessels as resistive wires, and the heart as a battery (Figure 1).



**Figure 1** Cardiovascular circuit model

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$$\Delta P = QR$$

**Equation 1**

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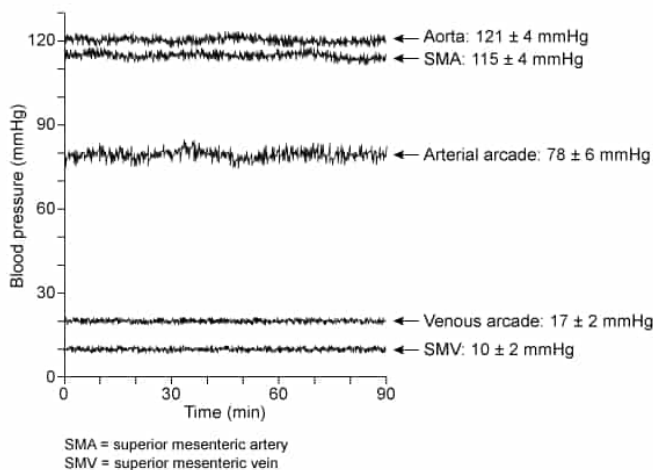
$$R = \frac{8\eta L}{\pi r^4}$$

**Equation 2**

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

In a study of the circulatory system in rats, researchers measured the intravascular blood pressure of the mesenteric blood vessels (the mesentery is the set of tissues that holds the intestines in place). Blood flows in the following order: aorta, superior mesenteric artery (SMA), arterial arcade, venous arcade, superior mesenteric vein (SMV).

For the procedure, catheters with pressure transducers were inserted into each vessel via laparotomy, an invasive procedure that involves a surgical incision into the abdominal cavity. Systolic blood pressure measurements were taken simultaneously at the proximal end (beginning) of each blood vessel.



**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

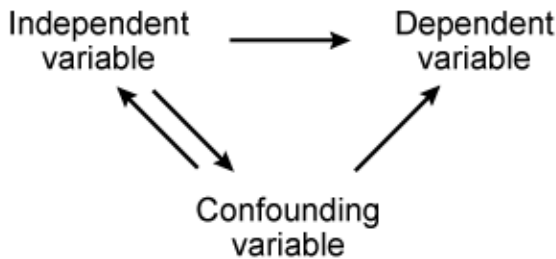
Which of the following experimental groups would aid in determining if the procedure described in the passage has a confounding variable?

- ☐ A. Group with laparotomy used without measuring blood pressure [9%]
- ☒ B. Group with minimally invasive aortic catheterization to measure mesenteric blood pressure [41%]
- ☐ C. Group fitted with noninvasive tail cuffs to measure systolic blood pressure [41%]
- ☐ D. Group with laparotomy to measure renovascular blood pressure [7%]

Correct

41% answered correctly

### Explanation:



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In an ideal experiment or study, any changes in the measured **dependent variable** are due to changes in the **independent variable**. However, this model may not be realistic due to **confounding variables**, uncontrolled variables that affect the dependent variable. If confounding variables cannot be eliminated, experiments should be designed to minimize their influence. To determine the magnitude of any effect of a possible confounding variable on the dependent variable, an experimental group that differs in the suspected confounding variable is tested.

In the study described in the passage, different blood vessels of rats (independent variable) were accessed through an invasive laparotomy procedure (suspected confounding variable) to measure mesenteric blood pressures (dependent variable). The inclusion of an alternative procedure that is minimally invasive or noninvasive but still measures mesenteric blood pressures would determine if laparotomy were a confounding variable by finding differences in the measured blood pressures.

**(Choices A and D)** The suspected confounding variable (procedural invasiveness of laparotomy) is not changed. In addition, the dependent variable (mesenteric blood pressure) should not change.

**(Choice C)** Although the suspected confounding variable (procedural invasiveness of laparotomy) is changed by using a tail cuff, the dependent variable (mesenteric blood pressures) should not change.

### Educational objective:

A confounding variable is an uncontrolled variable different from the independent variable but that still has an impact on the dependent variable.

The effect of a confounding variable can be observed by including a group in an experiment that differs in the confounding variable.

**Time spent:**0 sec

**Subject:**

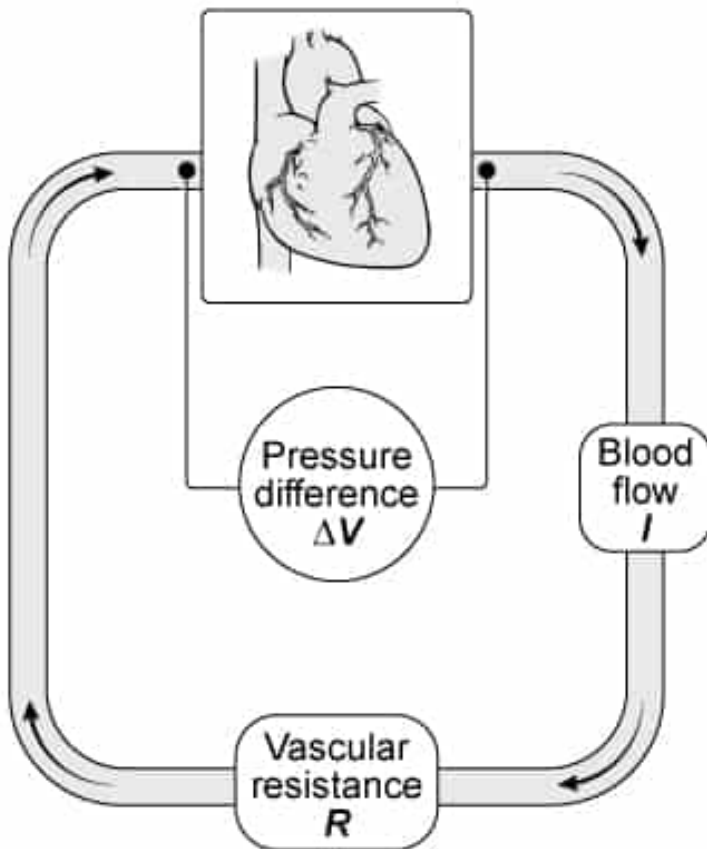
Physics

**QID:**400217

**System:**

Electrostatics and Circuits

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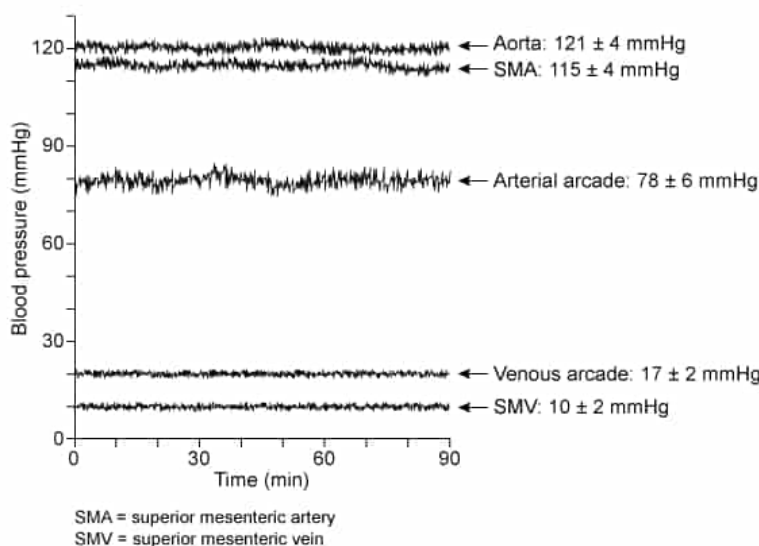
$$R = \frac{8\eta L}{\pi r^4}$$

## Equation 2

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

In a study of the circulatory system in rats, researchers measured the intravascular blood pressure of the mesenteric blood vessels (the mesentery is the set of tissues that holds the intestines in place). Blood flows in the following order: aorta, superior mesenteric artery (SMA), arterial arcade, venous arcade, superior mesenteric vein (SMV).

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**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

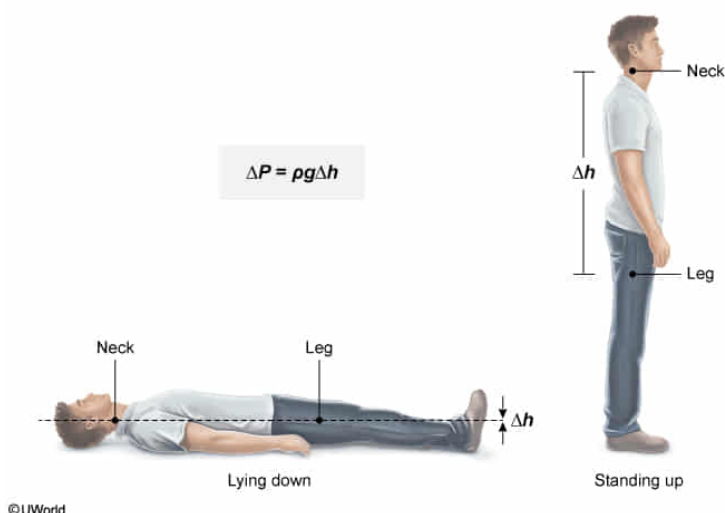
The blood pressures of an artery in the neck and an artery in the leg of a person lying down are measured, and their difference calculated. When the blood pressures are taken after the person stands up, their difference:

- ☐ A. decreases because flow resistance is greater over horizontal distances. [6%]
- ☐ B. remains the same because blood is modeled as an ideal fluid. [4%]
- ☐ C. increases because viscous pressure acts in the direction of gravity. [42%]
- ✓ ☒ D. increases due to the hydrostatic pressure difference between the two locations. [46%]

Correct

45% answered correctly

### Explanation:



At any point, the pressure exerted by the weight (potential energy)

of a fluid is directly proportional to its depth; this pressure is known as **hydrostatic pressure**. The hydrostatic pressure difference  $\Delta P$  between any two points in a fluid is found by

$$\Delta P = \rho g \Delta h$$

where  $\rho$  is the density of the fluid,  $g$  is the acceleration due to gravity, and  $\Delta h$  is the vertical distance between the locations where the initial and final pressures are measured.

When the person is lying down, the neck and the leg are at about the same height, making  $\Delta h$  negligible. However, when the person stands up, the  $\Delta h$  between the neck and the leg increases significantly. The blood pressure difference increases due to the hydrostatic pressure of the column of blood from above the leg to below the neck.

**(Choice A)** From Equation 2, flow resistance is due to the viscosity of blood and the geometry of the vessel through which it flows. Flow resistance is independent of the direction of flow.

**(Choice B)** An ideal fluid is one that possesses negligible viscosity, and an ideal fluid would still experience hydrostatic pressure. In addition, blood is not modeled here as an ideal fluid.

**(Choice C)** Viscosity is a measure of a fluid's resistance to flow due to internal frictional forces; it is not a measure of pressure. In addition, viscosity acts in the direction opposite the flow.

### **Educational objective:**

Hydrostatic pressure is the pressure exerted by the weight (potential energy) of a fluid. The hydrostatic pressure difference between two points in a fluid is proportional to the fluid's density, the gravitational acceleration, and the vertical distance between the two points.



**Time spent:**0 sec

**Subject:**

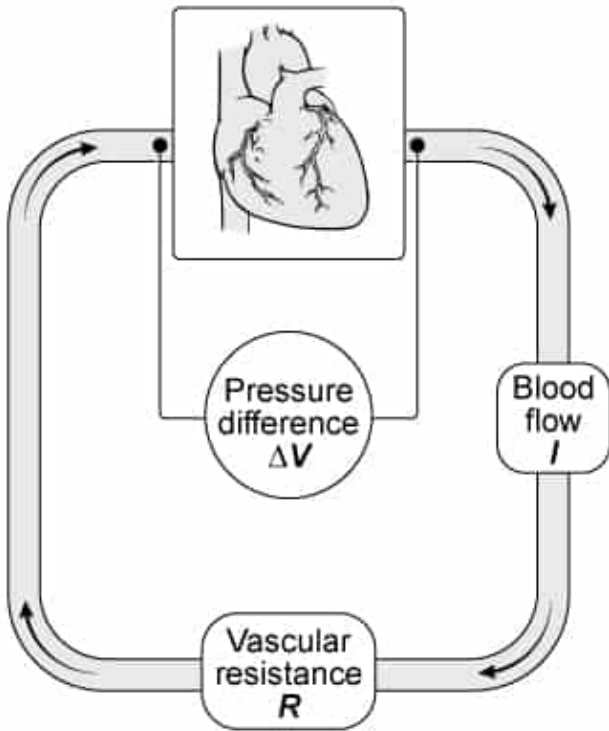
Physics

**QID:**400218

**System:**

Electrostatics and Circuits

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$$\Delta P = QR$$

**Equation 1**

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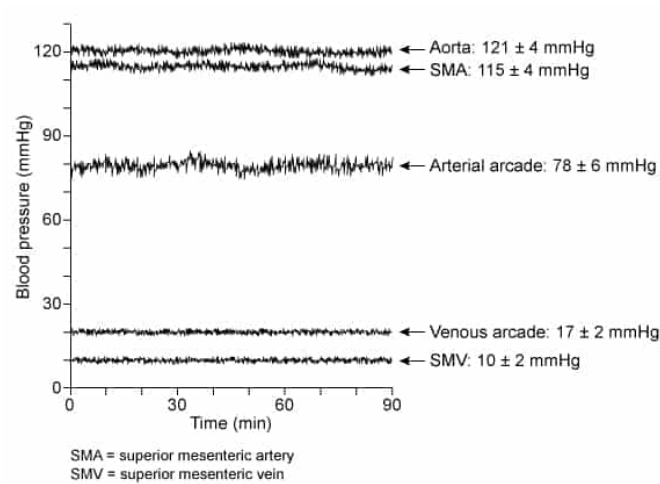
$$R = \frac{8\eta L}{\pi r^4}$$

**Equation 2**

where  $L$  is the length of the vessel, and  $r$  is its inner radius. If Equation 2 is combined with Equation 1, the resulting equation is Poiseuille's law.

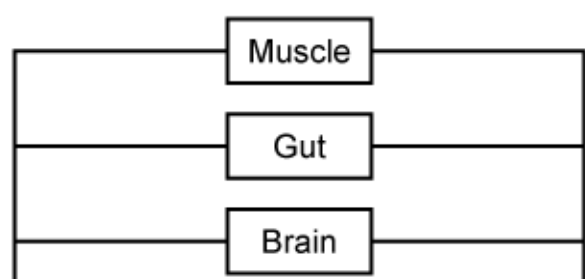
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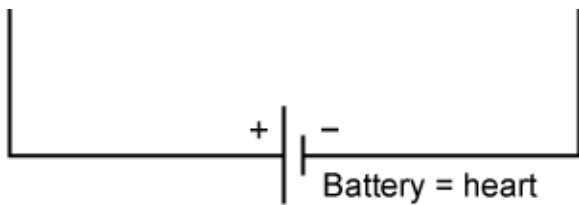
For the procedure, catheters with pressure transducers were inserted into each vessel via laparotomy, an invasive procedure that involves a surgical incision into the abdominal cavity. Systolic blood pressure measurements were taken simultaneously at the proximal end (beginning) of each blood vessel.



**Figure 2** Mesenteric blood pressure profile of rats (Note: SMA = superior mesenteric artery; SMV = superior mesenteric vein)

In the image shown, a simplified cardiovascular system is modeled as an electrical circuit. Which of the following would occur if the blood flow to the brain is blocked? (Assume that the blood pressure supplied by the heart remains unchanged.)



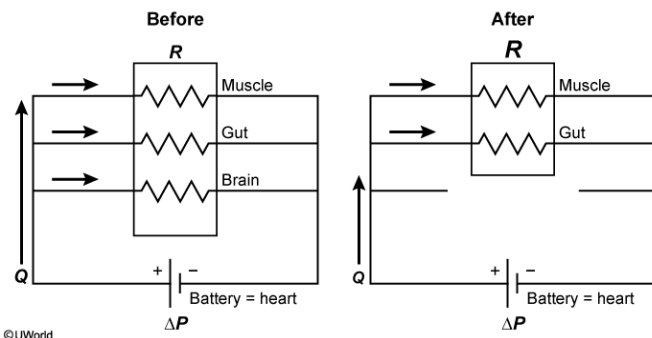


- ✓ ☒ A. Total vascular flow resistance will increase [38%]
- ☐ B. Pressure drop across the muscle will increase [10%]
- ☐ C. Blood flow through the gut will increase [48%]
- ☐ D. Blood flow throughout the cardiovascular system will stop [3%]

Correct

37% answered correctly

### Explanation:



The three biological components (resistors) shown in this model make a [parallel circuit](#), and the heart (battery) generates the pressure difference (voltage) that drives blood flow (current). The properties of parallel resistors are:

- The voltage drop across each resistor is the same. If they were connected in parallel with the battery, the voltage drop across each resistor would be equal to that of the voltage generated by the battery.
- The equivalent resistance decreases if a resistor is added in parallel. Conversely, the equivalent resistance increases if a resistor in parallel is removed.
- The current through each resistor is independent of that through the other resistors.
- The total current is equal to the sum of the currents through each component.

From the analogous properties, the equivalent resistance (total vascular flow resistance) will increase with the removal of a resistor (the brain) because it is connected in parallel with the other components.

**(Choice B)** The voltage drop across each resistor is equal to that of the battery, which does not change.

**(Choice C)** The current through each resistor in a parallel circuit is related only to the intrinsic resistance of the resistor and the electric potential (voltage). The vascular resistance of the gut and systemic blood pressure both remain constant after obstructing blood flow to the brain, so blood flow through the gut would not change.

**(Choice D)** If this were a **series circuit** in which the current flows through only one path, all flow would stop by blocking one element. However, parallel circuits have more than one path through which current can flow.

**Educational objective:**

For resistors connected in parallel, the voltage drop across each resistor is the same and the equivalent resistance increases if a resistor is removed. The current through each resistor in parallel is independent of the others, and the sum of each component current equals the total current.

**Time spent:**0 sec

**Subject:**  
Physics

**QID:**400219

**System:**  
Electrostatics and Circuits